EXTENSION WORK

During the process, the zinc loses electrons (is oxidised) to form zinc ions.

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

These electrons flow into the iron. Any iron atom which has lost electrons to form an ion immediately regains them. If the iron can't form ions, it can't rust.

to expose the iron. This is because the zinc is more reactive than iron, and so reacts with oxygen/water more readily than the iron does. The zinc corrodes instead of the iron.



▲ Figure 14.18 Galvanised iron doesn't rust even in constant contact with air and water.

PREVENTING RUSTING BY USING SACRIFICIAL PROTECTION



▲ Figure 14.19 A sacrificial anode on a ship

Zinc, magnesium or aluminium blocks are attached to metal hulls or keels of ships to prevent the iron/steel from rusting. For this to work, you have to use a metal that is *more reactive than iron;* the more reactive metal reacts (is oxidised) more readily in the presence of oxygen/water than the iron. The corrosion of the more reactive metal prevents the iron from rusting. Such blocks are called sacrificial anodes.

These sacrificial anodes have to be replaced occasionally when all the more reactive metal has been oxidised. This type of protection is used on large structures where it would be very difficult to use a barrier method effectively.

Galvanising is a combination of a barrier method and **sacrificial protection**.

Underground pipelines are also protected using sacrificial anodes. In this case, lumps of magnesium are attached at intervals along the pipe. The very reactive magnesium corrodes in preference to the iron. The electrons produced as the magnesium forms its ions prevent the ionisation of the iron.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING (6)

1 a List the following metals in order of decreasing reactivity: aluminium, copper, iron, sodium.

b Some magnesium powder was mixed with some copper(II) oxide and heated strongly. There was a vigorous reaction, producing a lot of sparks and a bright flash of light.

(5) i State the names of the products of the reaction.

SKILLS PROBLEM SOLVING (9) ii Write a balanced symbol equation for the reaction.

SKILLS REASONING iii Explain which substance in the reaction has been reduced.

iv Explain which substance is the oxidising agent.

c If a mixture of zinc powder and cobalt(II) oxide is heated, the following reaction occurs:

$$Zn(s) + CoO(s) \rightarrow ZnO(s) + Co(s)$$

(7) i Explain which metal is higher in the reactivity series.

SKILLS CRITICAL THINKING (9) ii The zinc can be described as a reducing agent. Using this example, explain what is meant by the term 'reducing agent'.

iii Explain which substance in this reaction has been oxidised.

SKILLS REASONING

8

d Aluminium, chromium and manganese are all moderately reactive metals. Use the following information to arrange them in the correct reactivity series order. Start with the most reactive one.

- Chromium is manufactured by heating chromium(III) oxide with aluminium.
- If manganese is heated with aluminium oxide there is no reaction.
- If manganese is heated with chromium(III) oxide, chromium is produced.

10

2 Study the following equations and in each case decide whether the substance in bold type is oxidised or reduced. Explain your choice in terms of either oxygen transfer or electron transfer as appropriate.

a **Zn(s)** + CuO(s) → ZnO(s) + Cu(s)

b **Fe_2O_3(s)** + 3C(s) → 2Fe(s) + 3CO(g)

c **Mg(s)** + Zn^{2+}(aq) → Mg^{2+}(aq) + Zn(s)

d Zn(s) + **Cu^{2+}(aq)** → Zn^{2+}(aq) + Cu(s)

8

3 The equation for the reaction when solid magnesium and solid lead(II) oxide are heated together is:

Mg(s)+ PbO(s) → MgO(s) + Pb(s)

Explain what this tells you about the position of lead in the reactivity series.

6

4 Some iron filings were shaken with some copper(II) sulfate solution. The ionic equation for the reaction is:

Fe(s) + Cu^{2+}(aq) → Fe^{2+}(aq) + Cu(s)

a Explain any one change that you would observe during this reaction.

9

b Explain which substance has been oxidised in this reaction.

SKILLS CRITICAL THINKING

c Write down the full (not ionic) equation for this reaction.

SKILLS REASONING

8

5 Some experiments were carried out to place the metals copper, nickel and silver in reactivity series order.

Experiment 1: A piece of copper was placed in some green nickel(II) sulfate solution. There was no change to either the copper or the solution.

Experiment 2: A coil of copper wire was suspended in some silver nitrate solution. A furry grey growth appeared on the copper wire, out of which grew spiky silvery crystals. The solution gradually turned from colourless to blue.

a Use this information to place copper, nickel and silver in reactivity series order, starting with the most reactive one.

b In another experiment, a piece of nickel was placed in some copper(II) sulfate solution.

6

i State one change that you would observe during this reaction.

SKILLS PROBLEM SOLVING

9

ii Write the full balanced equation for this reaction.

iii Write the ionic equation for this reaction and use it to explain which substance has been oxidised during the reaction.

SKILLS REASONING

8

6 a Look carefully at the following equations and then explain what you can say about the position of the metal X in the reactivity series.

X(s) + 2HCl(aq) → XCl_2(aq) + H_2(g)

X(s) + $CuSO_4$(aq) → XSO_4(aq) + Cu(s)

X(s) + $FeSO_4$(aq) → no reaction

SKILLS REASONING 8

b Decide whether X will react with the following substances. If it will react, state the names of the products, if not state 'no reaction'.

i Silver nitrate solution
ii Zinc oxide
iii Cold water
iv Copper(II) chloride solution
v Dilute sulfuric acid

SKILLS CRITICAL THINKING 5

7 If you add some powdered aluminium to a small amount of cold dilute hydrochloric acid in a boiling tube, very little happens. If you warm this gently, it starts to fizz very rapidly.

a State the name of the gas given off to produce the fizzing.

b If you used an excess of hydrochloric acid, the result would be a colourless solution. State the name of the solution.

SKILLS PROBLEM SOLVING 9

c Write the full balanced equation for the reaction.

SKILLS REASONING 6

d Explain why the aluminium hardly reacts at all with the dilute acid in the cold, but reacts vigorously after even gentle heating.

SKILLS EXECUTIVE FUNCTION 8

8 If you have some small pieces of the metal titanium and any simple apparatus that you might need, describe how you would find out the approximate position of titanium in the reactivity series, using only water and dilute hydrochloric acid. You only need to find out that the reactivity is 'similar to iron' or 'similar to magnesium', for example. Your experiments should be done in an order that guarantees maximum safety. For example, if titanium's reactivity turned out to be similar to potassium, dropping it into dilute hydrochloric acid wouldn't be a good idea!

SKILLS CRITICAL THINKING 5

9 In the past, cars were made from mild steel, which was then painted. In more modern cars, the mild steel is galvanised before it is painted.

a State the conditions under which iron/steel rusts.

6

b Explain how painting prevents iron/steel from rusting.

c What is meant by *galvanised steel*?

7

d Describe and explain the effect that galvanised steel has on the life of the car.

SKILLS INTERPRETATION 8

10 A student tested four unknown metals (the names are all imaginary) with different metal salt solutions. She put a tick (✔) if there was a reaction and a cross (✖) if there was no change. Some of her results are shown in the table.

	Pearsonium chloride solution	Mollium chloride solution	Rosium chloride solution	Amelium chloride solution
pearsonium	✖		✖	
mollium	✔	✖	✔	✔
rosium			✖	✖
amelium				✖

a Copy and complete the table.

SKILLS CRITICAL THINKING

b Arrange the metals in an order of reactivity (most reactive first).

SKILLS PROBLEM SOLVING 6

c Copy and complete the equation:

mollium + pearsonium chloride →